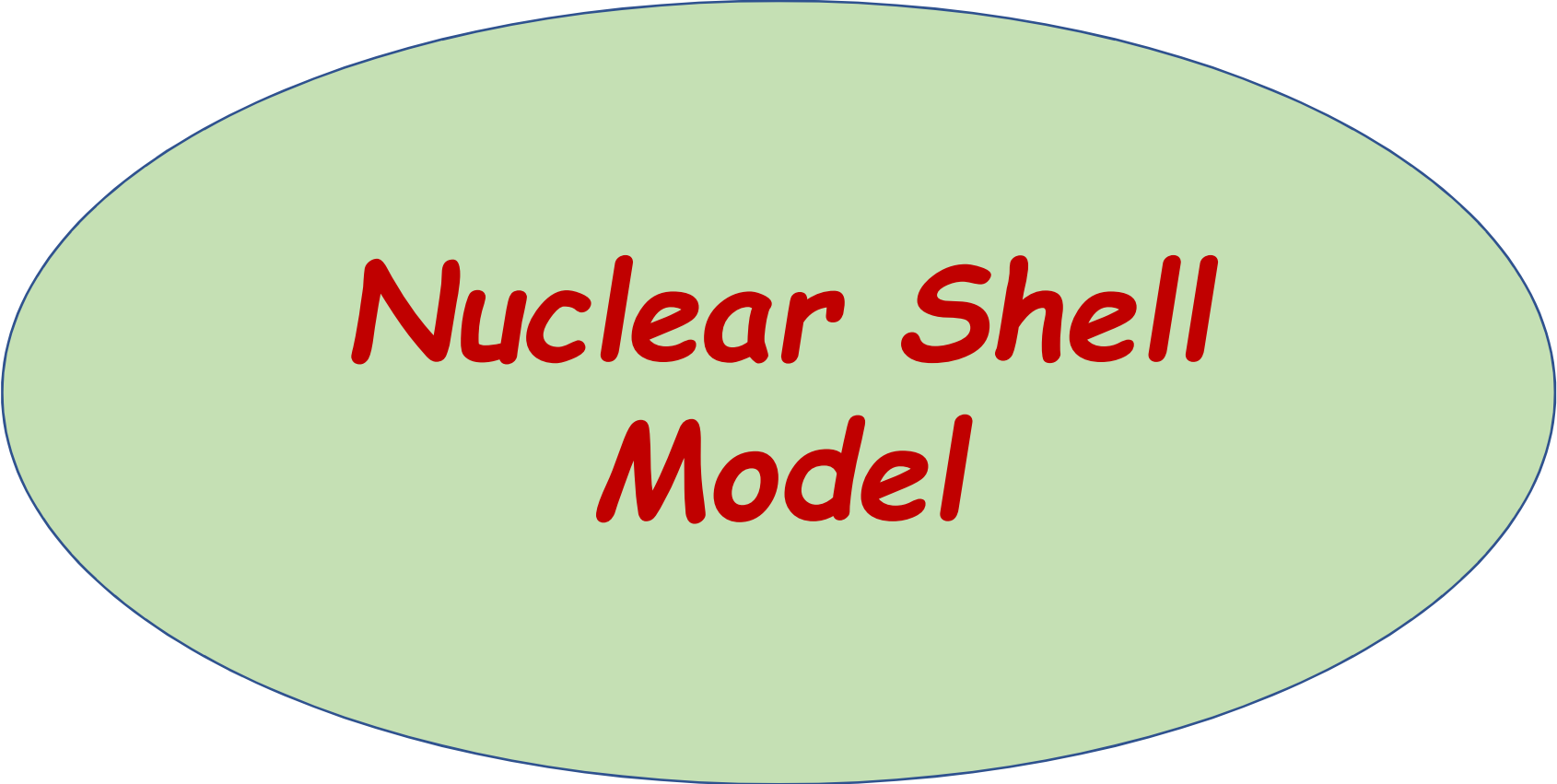


Contd.



Nuclear Shell Model

Extreme Single-Particle Shell model

3. Nuclear magnetic moment: **even-even nuclei**

- *Nucleons are paired up in each energy level*
- *Paired nucleons give zero spin and zero magnetic moment*
- *All e-e nuclei in their ground state have*

$$I^\pi = 0^+$$

- *Each pair contributes total magnetic moment = 0*
- *Total magnetic moment for the ground state of e-e nuclei*

$$\langle \mu_z \rangle = 0$$

- *This is consistent with the experimental observation*
- *For the excited states, pairs break giving rise to finite $\langle \mu_z \rangle$*

Extreme Single-Particle Shell model

3. Nuclear magnetic moment: **Odd-A nuclei**

- *There is one neutron or proton that is unpaired*
- *This last un-paired nucleon decides the properties of the nucleus*
- *We should calculate $\langle \mu_z \rangle$ for the last unpaired nucleon*

$$\mu_z = (g_l L_z + g_s S_z) \frac{\mu_N}{\hbar}$$

For the last (unpaired) nucleon, $\vec{J} = \vec{L} + \vec{S}$

$$J_z = L_z + S_z \quad m_j = m_l + m_s$$

Quantum state defined by $|l, s, j, m_j\rangle$

Measured value is generally the average value

So, we should calculate the average value

$$\langle \mu_z \rangle = \langle j, m_j = j | g_l L_z + g_s S_z | j, m_j = j \rangle \frac{\mu_N}{\hbar}$$

$$\mu_N = \frac{e\hbar}{2m_p}$$

$$\mu_N = 3.1525 \times 10^{-8} \text{ eV/T}$$

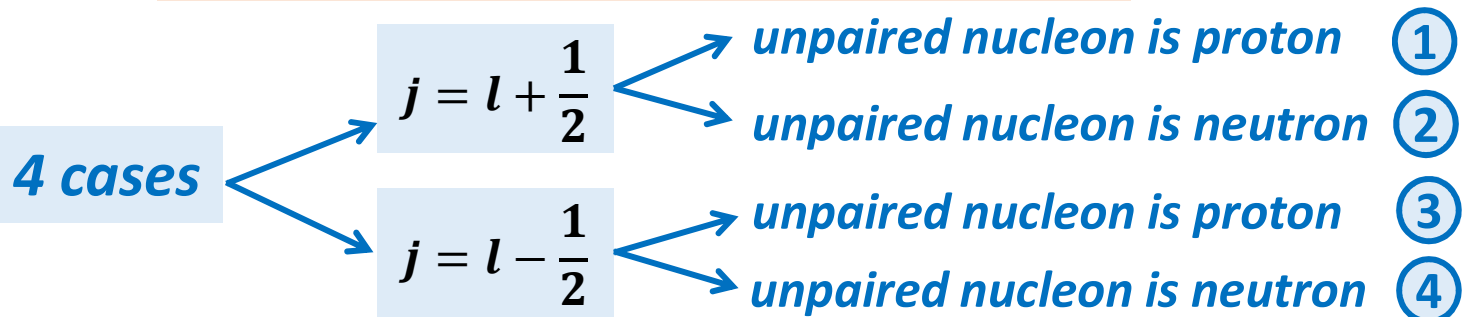
$$g_s = -3.8260837 \text{ for n} \\ = 5.5856912 \text{ for p}$$

$$g_l = 1 \text{ for p} \\ = 0 \text{ for n}$$

Extreme Single-Particle Shell model

3. Nuclear magnetic moment

$$\langle \mu_z \rangle = \langle j, m_j = j | g_l L_z + g_s S_z | j, m_j = j \rangle \frac{\mu_N}{\hbar}$$



Case ①, ②

$$j = l + \frac{1}{2}$$

$$m_j = m_l + m_s \Rightarrow l + \frac{1}{2} = m_l + m_s$$

$$\langle L_z \rangle = l\hbar$$

$$\langle S_z \rangle = \hbar/2$$

$$m_l = l, l-1, \dots, -l$$

$$m_s = \frac{1}{2}, -\frac{1}{2}$$

$$\langle \mu_z \rangle = \left(g_l l\hbar + g_s \frac{\hbar}{2} \right) \frac{\mu_N}{\hbar} = \left(g_l l + \frac{g_s}{2} \right) \mu_N = \left[g_l \left(j - \frac{1}{2} \right) + \frac{g_s}{2} \right] \mu_N$$

Case - ① proton

$$\langle \mu_z \rangle = \left[\left(j - \frac{1}{2} \right) + \frac{g_s}{2} \right] \mu_N = (j + 2.2928) \mu_N$$

Case - ② neutron

$$\langle \mu_z \rangle = \frac{g_s}{2} \mu_N = -1.913 \mu_N$$

$$g_s = 5.5856912 \text{ for } p \\ = -3.8260837 \text{ for } n$$

Extreme Single-Particle Shell model

3. Nuclear magnetic moment

Case ③, ④ $j = l - \frac{1}{2}$

$$\langle \mu_z \rangle = \langle j, m_j = j | g_l L_z + g_s S_z | j, m_j = j \rangle \frac{\mu_N}{\hbar}$$

$$m_j = m_l + m_s \Rightarrow l - \frac{1}{2} = m_l + m_s \quad m_l = l, l-1, \dots, -l$$

$$m_s = \frac{1}{2}, -\frac{1}{2}$$

$$m_l + m_s = l - \frac{1}{2} \text{ for 2 combinations of } m_l \text{ and } m_s$$

$$m_l = l, m_s = -\frac{1}{2}$$

and

$$m_l = l-1, m_s = \frac{1}{2}$$

Mixed states

$$|j, m_j = j\rangle = a\psi_1 + b\psi_2$$

$$\psi_1 = |m_l = l, m_s = -1/2\rangle$$

$$\psi_2 = |m_l = l-1, m_s = 1/2\rangle$$

$$\langle \mu_z \rangle = \langle a\psi_1 + b\psi_2 | \mu_z | a\psi_1 + b\psi_2 \rangle$$

$$\langle \mu_z \rangle = \langle a\psi_1 | \mu_z | a\psi_1 \rangle + \langle b\psi_2 | \mu_z | b\psi_2 \rangle + \langle a\psi_1 | \mu_z | b\psi_2 \rangle + \langle b\psi_2 | \mu_z | a\psi_1 \rangle$$

$$\langle \mu_z \rangle = |a|^2 \langle \psi_1 | \mu_z | \psi_1 \rangle + |b|^2 \langle \psi_2 | \mu_z | \psi_2 \rangle \quad \textcircled{1}$$

ψ_1, ψ_2 are orthogonal

$$\langle \psi_1 | \mu_z | \psi_1 \rangle = \langle m_l = l, m_s = -1/2 | \mu_z | m_l = l, m_s = -1/2 \rangle \quad \textcircled{2}$$

$$\langle \psi_2 | \mu_z | \psi_2 \rangle = \langle m_l = l-1, m_s = 1/2 | \mu_z | m_l = l-1, m_s = 1/2 \rangle \quad \textcircled{3}$$

Extreme Single-Particle Shell model

3. Nuclear magnetic moment

Case ③, ④ $j = l - \frac{1}{2}$

$$\langle \mu_z \rangle = \langle j, m_j = j | g_l L_z + g_s S_z | j, m_j = j \rangle \frac{\mu_N}{\hbar}$$

$$\langle \mu_z \rangle = |a|^2 \langle \psi_1 | \mu_z | \psi_1 \rangle + |b|^2 \langle \psi_2 | \mu_z | \psi_2 \rangle \quad \textcircled{1}$$

$$\langle \psi_1 | \mu_z | \psi_1 \rangle = \langle m_l = l, m_s = -1/2 | \mu_z | m_l = l, m_s = -1/2 \rangle \quad \textcircled{2}$$

$$\langle \psi_2 | \mu_z | \psi_2 \rangle = \langle m_l = l - 1, m_s = 1/2 | \mu_z | m_l = l - 1, m_s = 1/2 \rangle \quad \textcircled{3}$$

Solve eq. 2, 3 and substitute in 1

$$\langle \psi_1 | \mu_z | \psi_1 \rangle = \langle m_l = l, m_s = -1/2 | g_l L_z + g_s S_z | m_l = l, m_s = -1/2 \rangle \frac{\mu_N}{\hbar}$$

$$\langle \psi_1 | \mu_z | \psi_1 \rangle = \left(g_l l \hbar - g_s \frac{\hbar}{2} \right) \frac{\mu_N}{\hbar} \quad \textcircled{2a}$$

$$\langle \psi_2 | \mu_z | \psi_2 \rangle = \langle m_l = l - 1, m_s = 1/2 | g_l L_z + g_s S_z | m_l = l - 1, m_s = 1/2 \rangle \frac{\mu_N}{\hbar}$$

$$\langle \psi_2 | \mu_z | \psi_2 \rangle = \left(g_l (l - 1) \hbar + g_s \frac{\hbar}{2} \right) \frac{\mu_N}{\hbar} \quad \textcircled{3a}$$

$$\langle \mu_z \rangle = |a|^2 \left(g_l l \hbar - g_s \frac{\hbar}{2} \right) \frac{\mu_N}{\hbar} + |b|^2 \left(g_l (l - 1) \hbar + g_s \frac{\hbar}{2} \right) \frac{\mu_N}{\hbar} \quad \textcircled{1a}$$

Extreme Single-Particle Shell model

3. Nuclear magnetic moment

Case ③, ④ $j = l - \frac{1}{2}$

$$\langle \mu_z \rangle = \langle j, m_j = j | g_l L_z + g_s S_z | j, m_j = j \rangle \frac{\mu_N}{\hbar}$$

$$\langle \mu_z \rangle = |a|^2 \left(g_l l - \frac{g_s}{2} \right) \mu_N + |b|^2 \left(g_l (l - 1) + \frac{g_s}{2} \right) \mu_N \quad 1a$$

$J_+ = J_x + iJ_y \Rightarrow$ Raising operator

$J_- = J_x - iJ_y \Rightarrow$ Lowering operator

$$J_+ |j, m_j\rangle = \left(\sqrt{(j - m_j)(j + m_j + 1)} \right) |j, m_j + 1\rangle$$

$$|j, m_j = j\rangle = a\psi_1 + b\psi_2$$

$$\psi_1 = |m_l = l, m_s = -1/2\rangle$$

$$\psi_2 = |m_l = l - 1, m_s = 1/2\rangle$$

$$|j, m_j = j\rangle = a|m_l = l, m_s = -1/2\rangle + b|m_l = l - 1, m_s = 1/2\rangle$$

$$J_+ [|j, m_j = j\rangle] = 0 = J_+ [a|m_l = l, m_s = -1/2\rangle + b|m_l = l - 1, m_s = 1/2\rangle]$$

$$(L_+ + S_+) [a|m_l = l, m_s = -1/2\rangle + b|m_l = l - 1, m_s = 1/2\rangle] = 0$$

$$a|m_l = l, m_s = 1/2\rangle + b\sqrt{2l}|m_l = l, m_s = 1/2\rangle = 0$$

$$a + b\sqrt{2l} = 0$$

$$a = -b\sqrt{2l}$$

$$\text{and } |a|^2 + |b|^2 = 1$$



$$|b|^2 = 1/(2l + 1)$$

$$|a|^2 = 2l/(2l + 1)$$

Substitute eq. 4 in eq. 1a

Extreme Single-Particle Shell model

3. Nuclear magnetic moment

Case (3), (4) $j = l - \frac{1}{2}$

$$\langle \mu_z \rangle = \langle j, m_j = j | g_l L_z + g_s S_z | j, m_j = j \rangle \frac{\mu_N}{\hbar}$$

$$\langle \mu_z \rangle = \frac{2l}{2l+1} \left(g_l l - \frac{g_s}{2} \right) \mu_N + \frac{1}{2l+1} \left(g_l(l-1) + \frac{g_s}{2} \right) \mu_N$$

1c

$$\langle \mu_z \rangle = \frac{\mu_N}{2l+1} \left[g_l(2l^2 + l - 1) + g_s \left(-l + \frac{1}{2} \right) \right]$$

$$\langle \mu_z \rangle = \frac{\mu_N}{2(j+1)} \left[g_l \left\{ 2 \left(j + \frac{1}{2} \right)^2 + \left(j + \frac{1}{2} \right) - 1 \right\} + g_s \left\{ - \left(j + \frac{1}{2} \right) + \frac{1}{2} \right\} \right]$$

$$\langle \mu_z \rangle = \frac{\mu_N j}{2(j+1)} [g_l(2j+3) - g_s]$$

1d

Gives the value of $\langle \mu_z \rangle$ for the ground state of odd-A nuclei

Case (3)
proton

$$\langle \mu_z \rangle = \frac{\mu_N j}{2(j+1)} [2j+3 - g_s]$$

$$\langle \mu_z \rangle = \frac{j}{j+1} [j - 1.2928] \mu_N$$

1e

Case (4)
neutron

$$\langle \mu_z \rangle = \frac{\mu_N j}{2(j+1)} [-g_s]$$

$$\langle \mu_z \rangle = \frac{j}{j+1} [1.9130] \mu_N$$

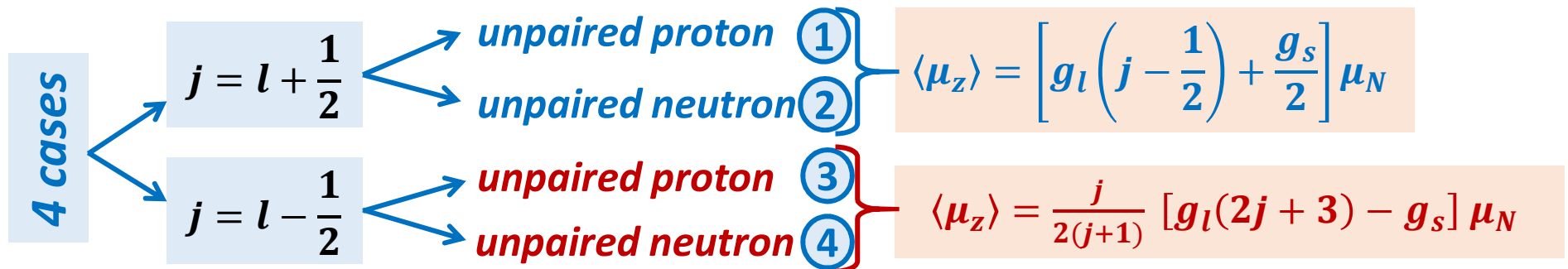
1f

Extreme Single-Particle Shell model

3. Nuclear magnetic moment

Summary

$$\langle \mu_z \rangle = \langle j, m_j = j | g_l L_z + g_s S_z | j, m_j = j \rangle \frac{\mu_N}{\hbar}$$



Case - ① proton	$j = l + \frac{1}{2}$	$\langle \mu_z \rangle = (j + 2.2928) \mu_N$
-----------------	-----------------------	--

Case - ② neutron	$j = l + \frac{1}{2}$	$\langle \mu_z \rangle = -1.913 \mu_N$
------------------	-----------------------	--

Case - ③ proton	$j = l - \frac{1}{2}$	$\langle \mu_z \rangle = \frac{j}{j+1} [j - 1.2928] \mu_N$
-----------------	-----------------------	--

Case - ④ neutron	$j = l - \frac{1}{2}$	$\langle \mu_z \rangle = \frac{j}{j+1} [1.9130] \mu_N$
------------------	-----------------------	--

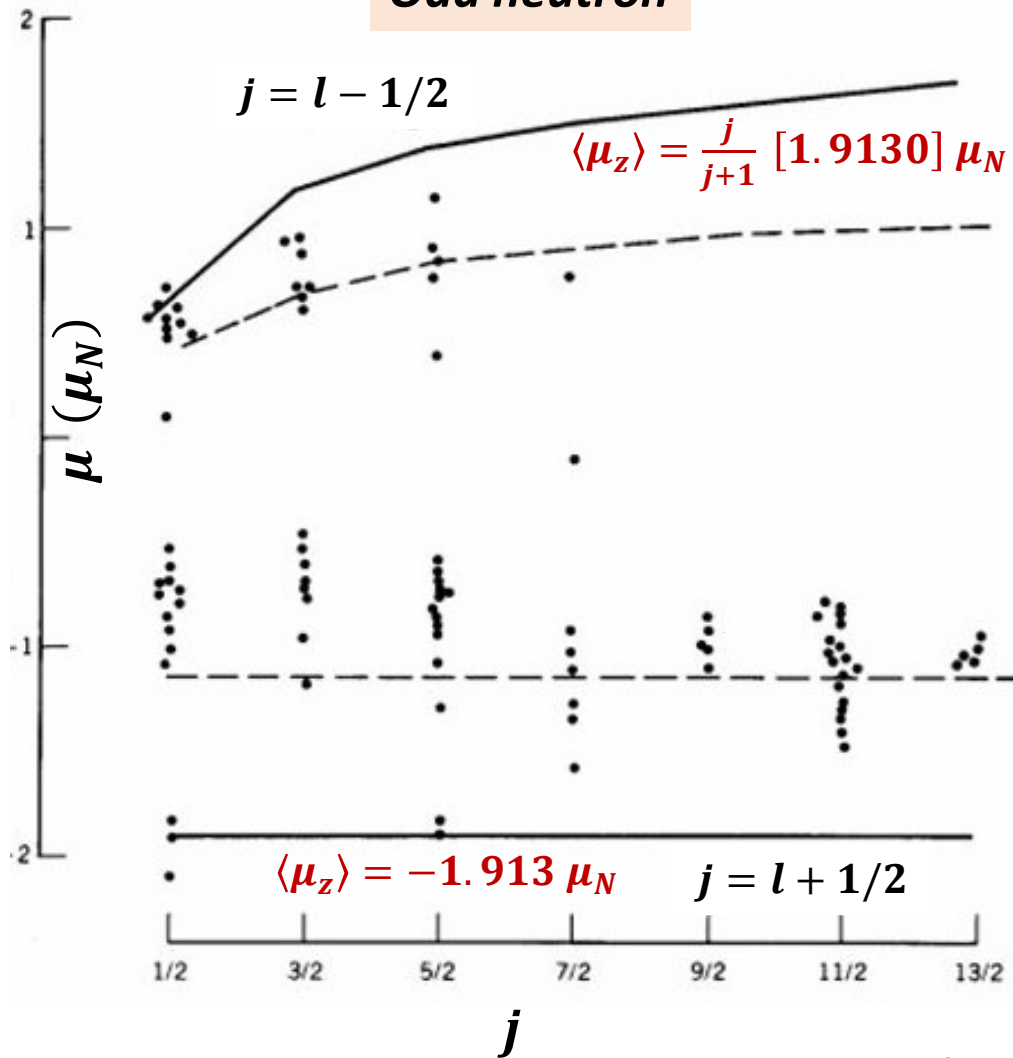
Schmidt limits for magnetic moments

Extreme Single-Particle Shell model

3. Nuclear magnetic moment

Summary

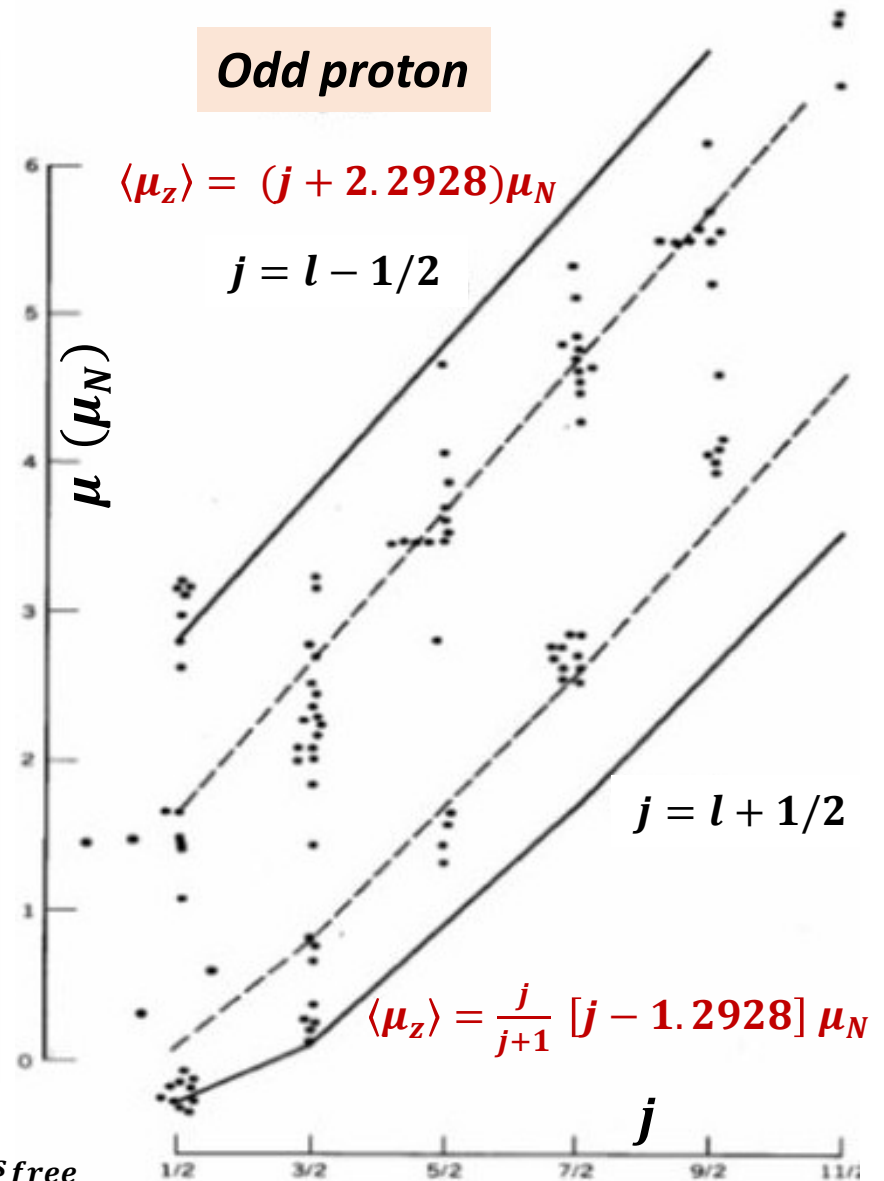
Odd neutron



Schmidt lines

Dashed lines: $g_s = 0.6 g_{sfree}$

Odd proton



Extreme Single-Particle Shell model

3. Nuclear magnetic moment

Summary

- ❑ *The experimental values are smaller in magnitude and have considerable scattering*
- ❑ *The experimental values fall within the limits of the Schmidt lines*
- ❑ *The Schmidt lines for magnetic moments obtained using the extreme single particle model gives a limit to the observed magnetic moments*
- ❑ *Defects in the theory:*
 - ❖ *Extreme single particle model is not purely the right description*
 - *More coupling between the paired neutrons/protons*
 - ❖ *Assumption that g_s for a nucleon in a nucleus is the same as g_s for a free nucleon*
 - *Meson cloud surround the nucleons*
 - *The meson cloud in nuclei where there are other surrounding nucleons differ from that for free nucleons*
 - *This may reduce the value of g_s for a nucleon in a nucleus*

Extreme Single-Particle Shell model

4. Electric Quadrupole Moment

$$Q = \int \Psi^* (3z^2 - r^2) \Psi d^3r = \int \Psi^* (3\cos^2\theta - 1) \Psi d^3r$$

❑ *The calculation of electric quadrupole moments in the shell model is done by evaluating the electric quadrupole operator, $3z^2 - r^2$, in a state in which the total angular momentum of the odd particle has its maximum projection along the z-axis (i.e. $m_j = +j$)*

❑ *Assuming odd particle : proton*

➤ *Ang. Mom, aligned with the z-axis $\Rightarrow$ proton is orbiting mostly in the x-y plane*

➤ *Negative quadrupole moment $Q \approx -\langle r^2 \rangle$*

➤ *More refined quantum mechanical calculations give*

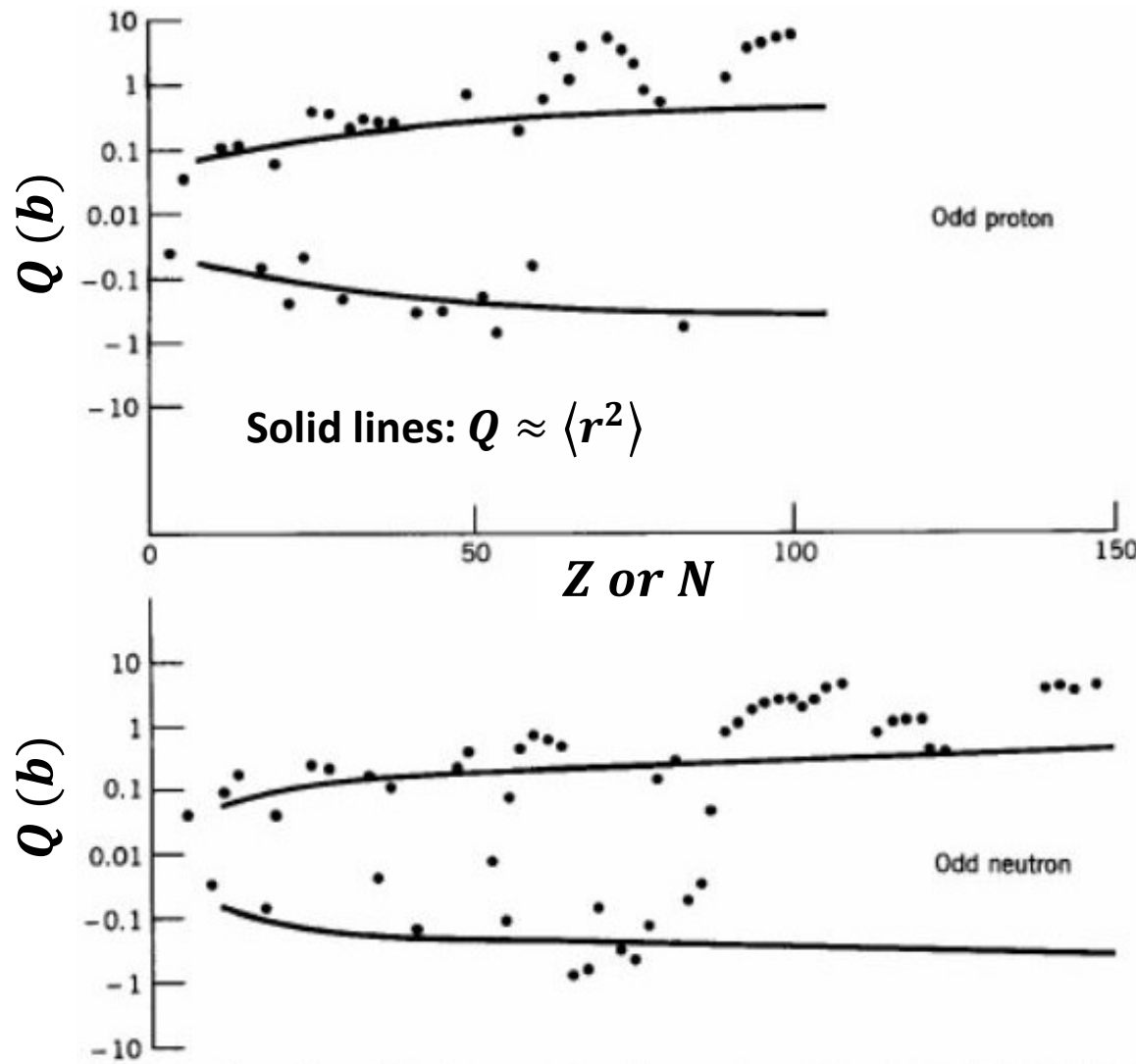
$$\langle Q_{sp} \rangle = -\frac{2j-1}{2(j+1)} \langle r^2 \rangle \quad \text{for an odd proton in a shell model state } j$$

If a subshell contains more than one particle

$$\langle Q \rangle = \langle Q_{sp} \rangle \left[1 - 2 \frac{n-1}{2j-1} \right]$$

Extreme Single-Particle Shell model

4. Electric Quadrupole Moment



The experimental data are within the limits,

except for regions

$60 < Z < 80$

$Z > 90$

$90 < N < 120$

$N > 140$

where the experimental values are more than an order of magnitude larger than predicted by the shell model

Schmidt limits for quadupole moments

Shell Model : Success and Limitations

- ❑ *Successful in predicting the ground state (and some excited states) spin-parity of nuclei*
- ❑ *Calculation of probability of making a transition from one state to another*
- ❑ *Gives justification of the asymmetry term in the Binding energy*
- ❑ *Partially successful in calculation of magnetic dipole moments*
- ❑ *Fails to predict the extremely large quadrupole moments of several barns observed for certain heavy nuclei*
- ❖ *Single particle Shell model cannot go too far in understanding all the properties of nuclei*
- ❖ *There are other aspects of nuclear structure that cannot be explained within the shell model*

Next step:

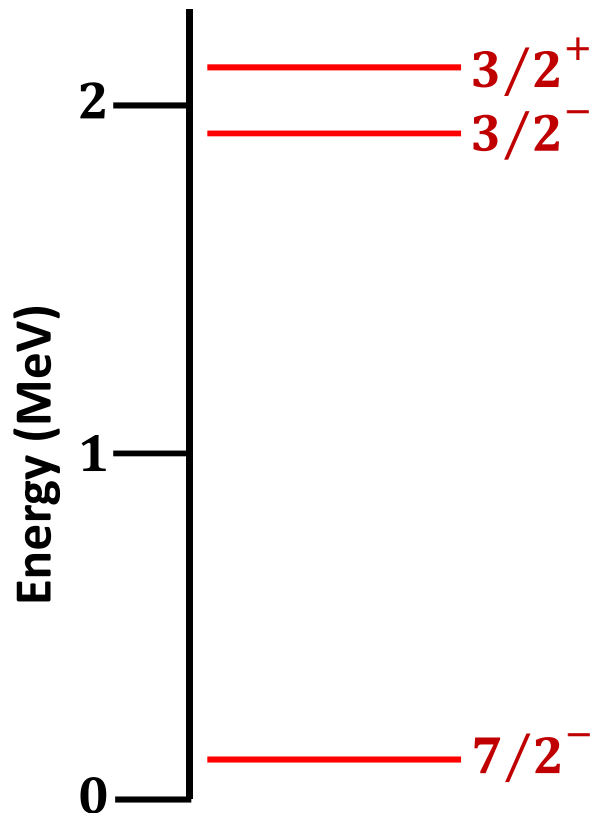
combining features of shell model and liquid drop model $\Rightarrow$ collective models

Shell Model → Collective structure

Odd-A nuclei

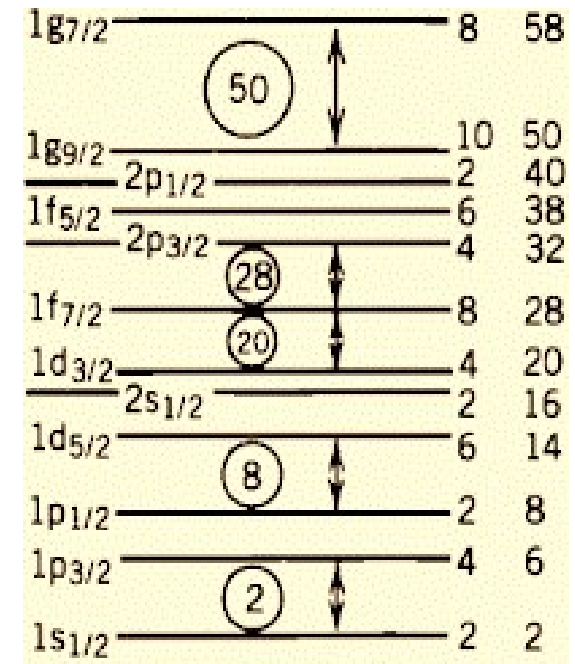
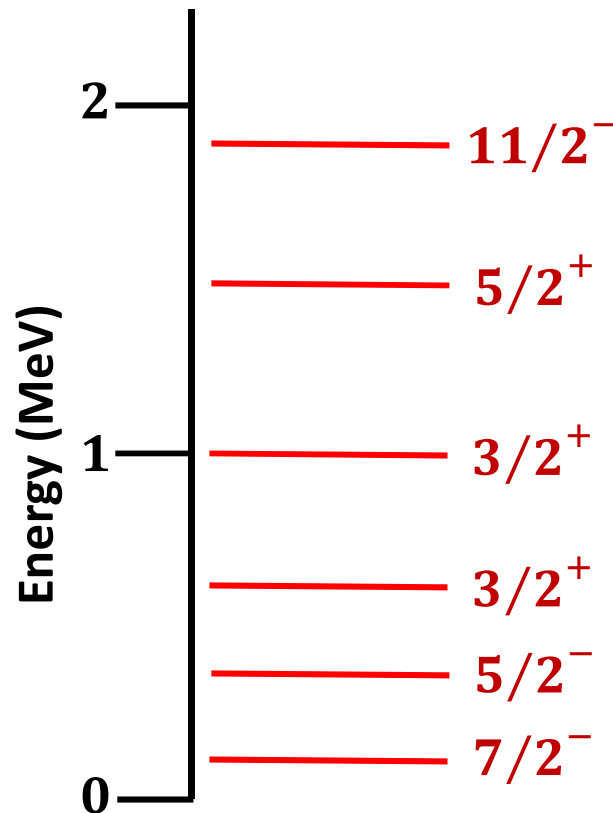
$^{41}_{20}\text{Ca}_{21}$

$Z = 20$
 $N = 21$



$^{43}_{20}\text{Ca}_{23}$

$Z = 20$
 $N = 23$

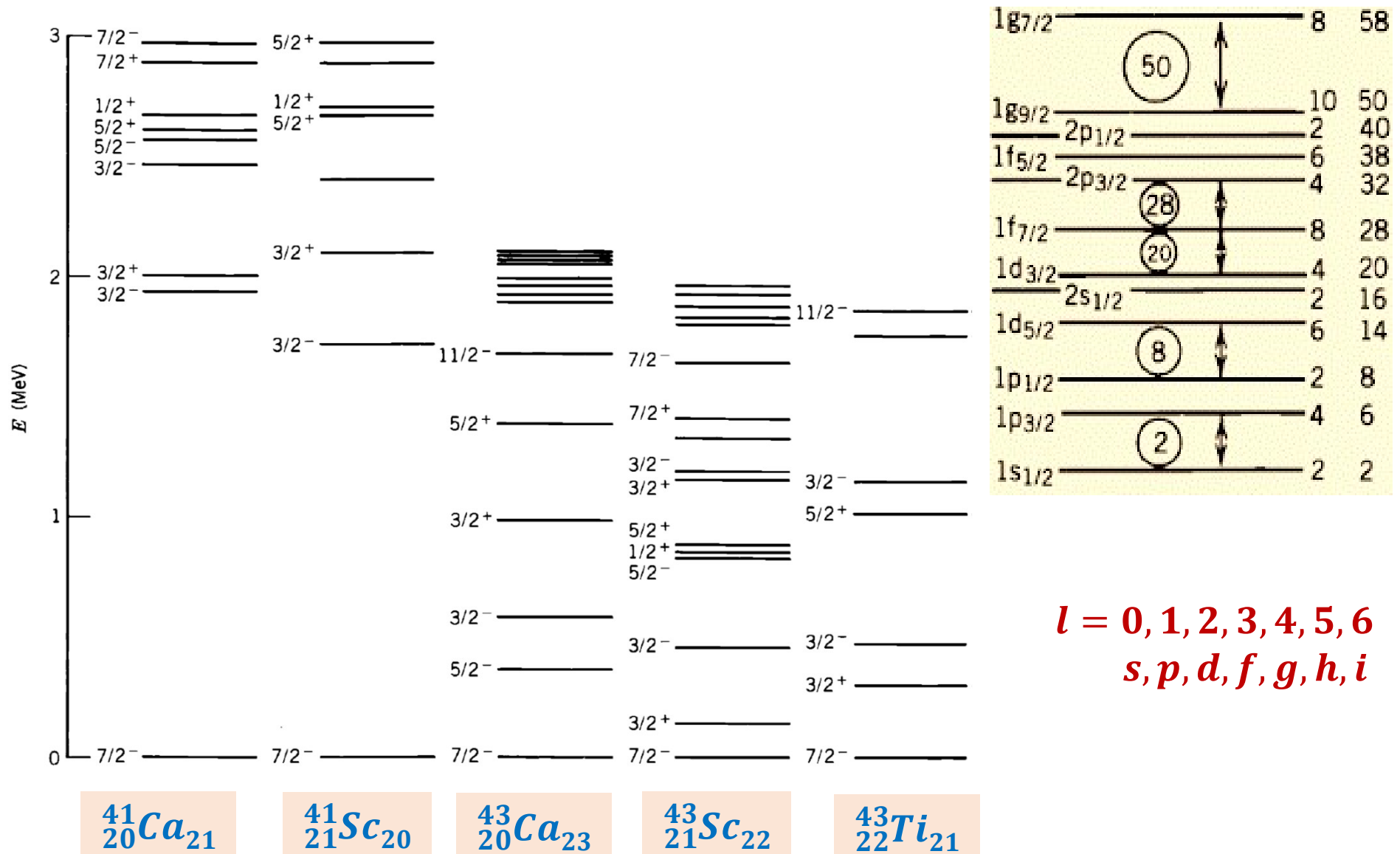


$l = 0, 1, 2, 3, 4, 5, 6$
 s, p, d, f, g, h, i

- ❑ In ^{43}Ca , there are several energy levels below 2 MeV
- ❑ Such low energies cannot be explained by shell model (breaking of nucleon pairs need ~ 2 MeV)

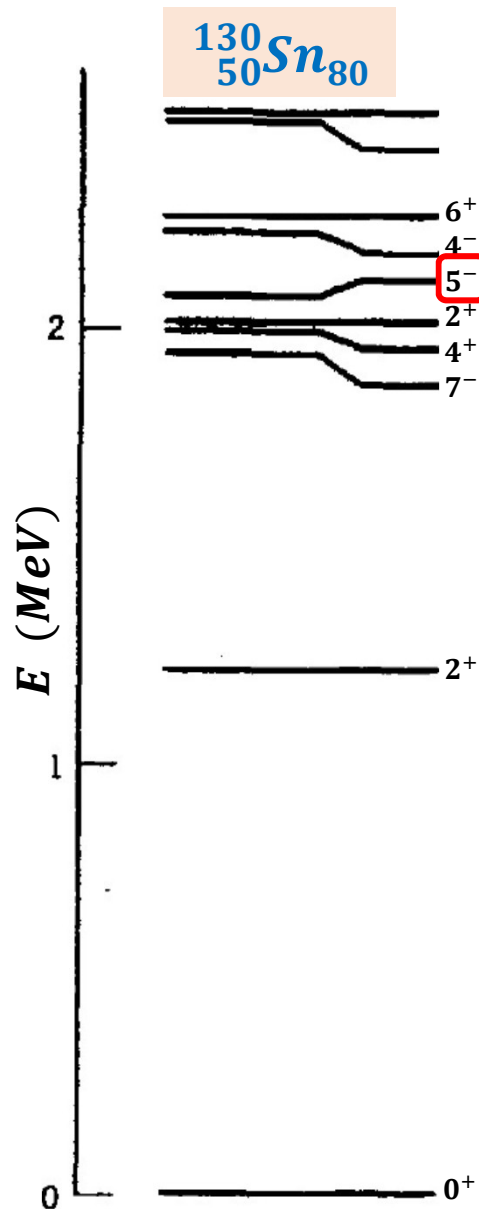
Shell Model → Collective structure

Odd-A nuclei



Shell Model → Collective structure

Even-even nuclei



Possibilities

Breaking of a pair in $3s_{1/2}$

Excitation of one n from $3s_{1/2}$ to $1h_{11/2}$

Ang. Mom.: $j_1 + j_2$ to $j_1 - j_2$

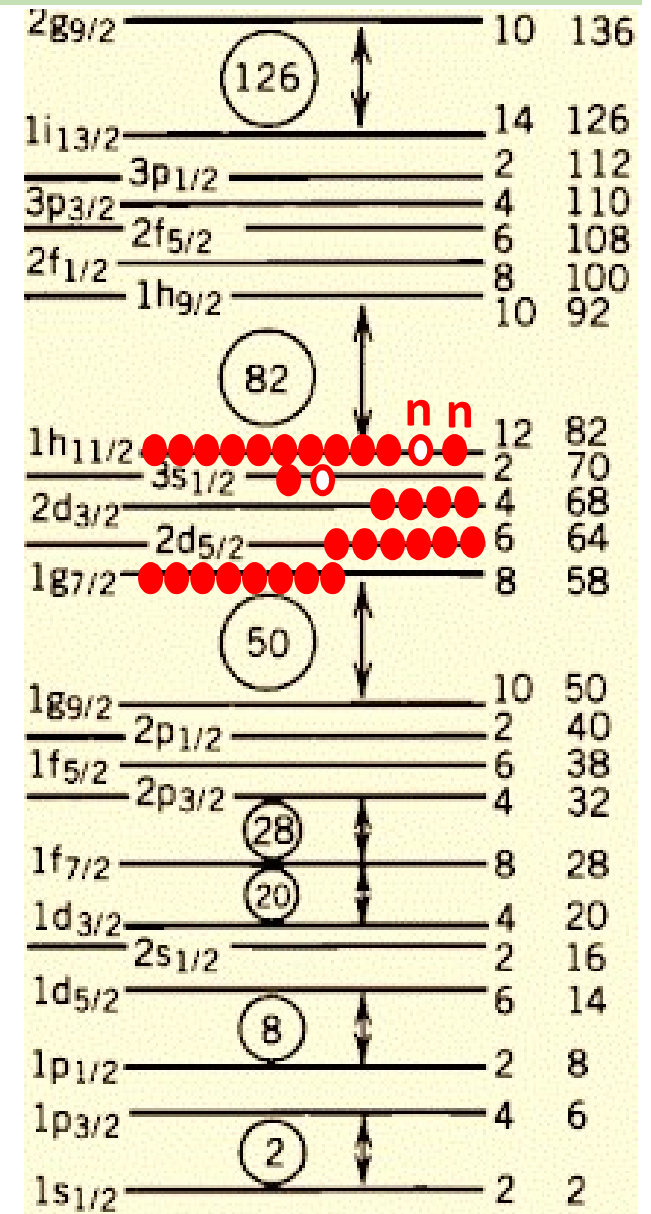
$$j_1 + j_2 = \frac{11}{2} + \frac{1}{2} = 6$$

$$j_1 - j_2 = \frac{11}{2} - \frac{1}{2} = 5$$

$$\pi = (-1^0)(-1^5) = -ve$$

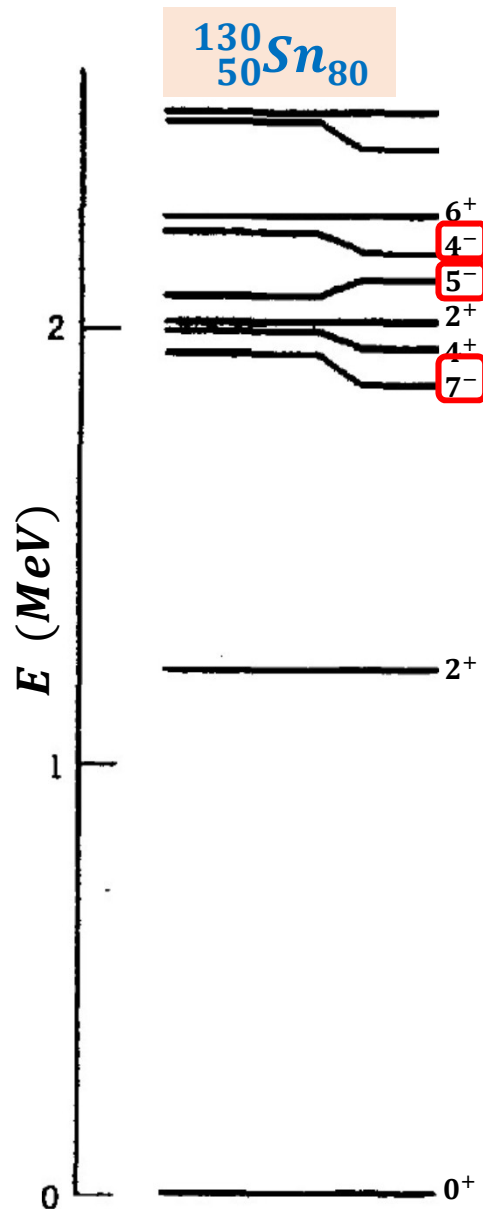
Possible states: 5^- , 6^-

$l = 0, 1, 2, 3, 4, 5, 6$
 s, p, d, f, g, h, i



Shell Model → Collective structure

Even-even nuclei



Possibilities

Breaking of a pair in $2d_{3/2}$

Excitation of one n from $2d_{3/2}$ to $1h_{11/2}$

Ang. Mom.: $j_1 + j_2$ to $j_1 - j_2$

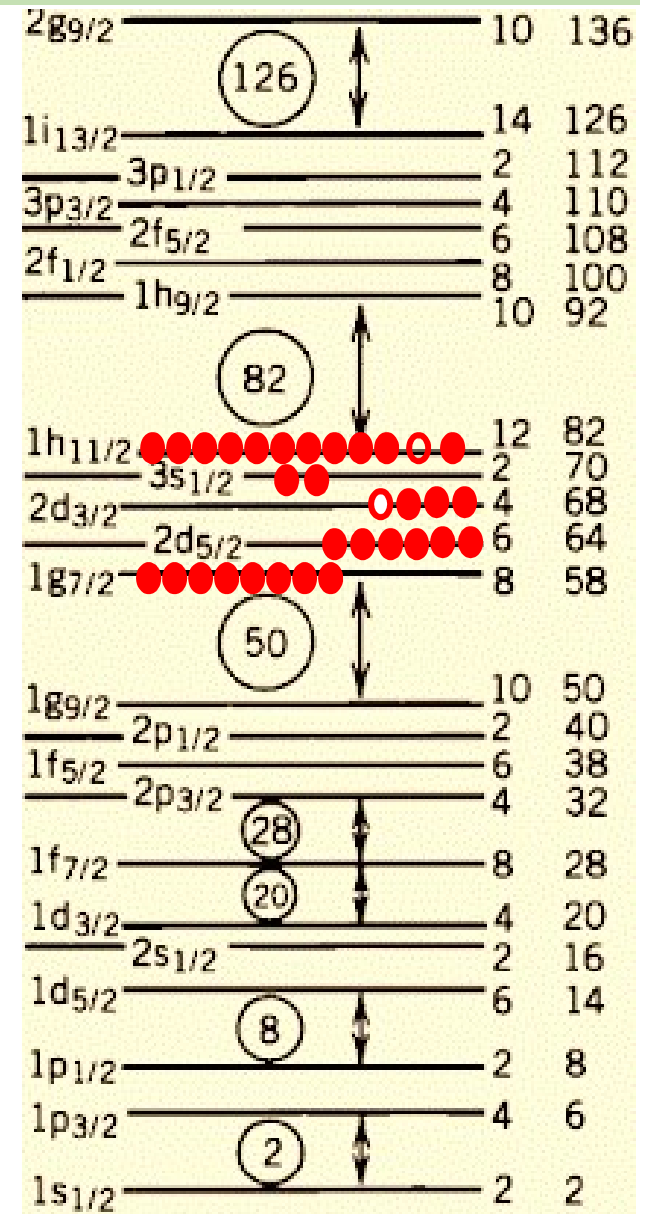
$$j_1 + j_2 = \frac{11}{2} + \frac{3}{2} = 7$$

$$j_1 - j_2 = \frac{11}{2} - \frac{3}{2} = 4$$

$$\pi = (-1^2)(-1^5) = -ve$$

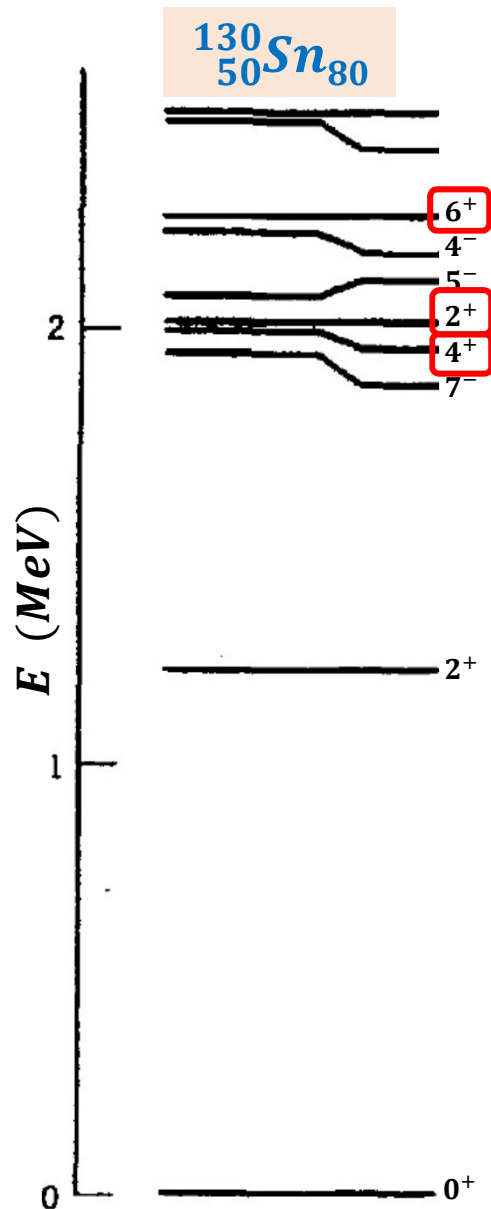
Possible states: 4^- , 5^- , 6^- , 7^-

$l = 0, 1, 2, 3, 4, 5, 6$
 s, p, d, f, g, h, i



Shell Model → Collective structure

Even-even nuclei



Possibilities

Breaking of a pair in $1h_{11/2}$

Coupling of the 2 unpaired neutrons in $1h_{11/2}$ giving ang. mom. > 0

Ang. Mom.: $j_1 + j_2$ to $j_1 - j_2$

$$j_1 + j_2 = \frac{11}{2} + \frac{11}{2} = 11$$

$$j_1 - j_2 = \frac{11}{2} - \frac{11}{2} = 0$$

$$\pi = (-1^5)(-1^5) = +ve$$

Possible states: $11^+, 10^+, \dots, 0^+$

2 particles in $1h_{11/2}$ are identical particles

⇒ Possible states: $10^+, 8^+, 6^+, 4^+, 2^+, 0^+$

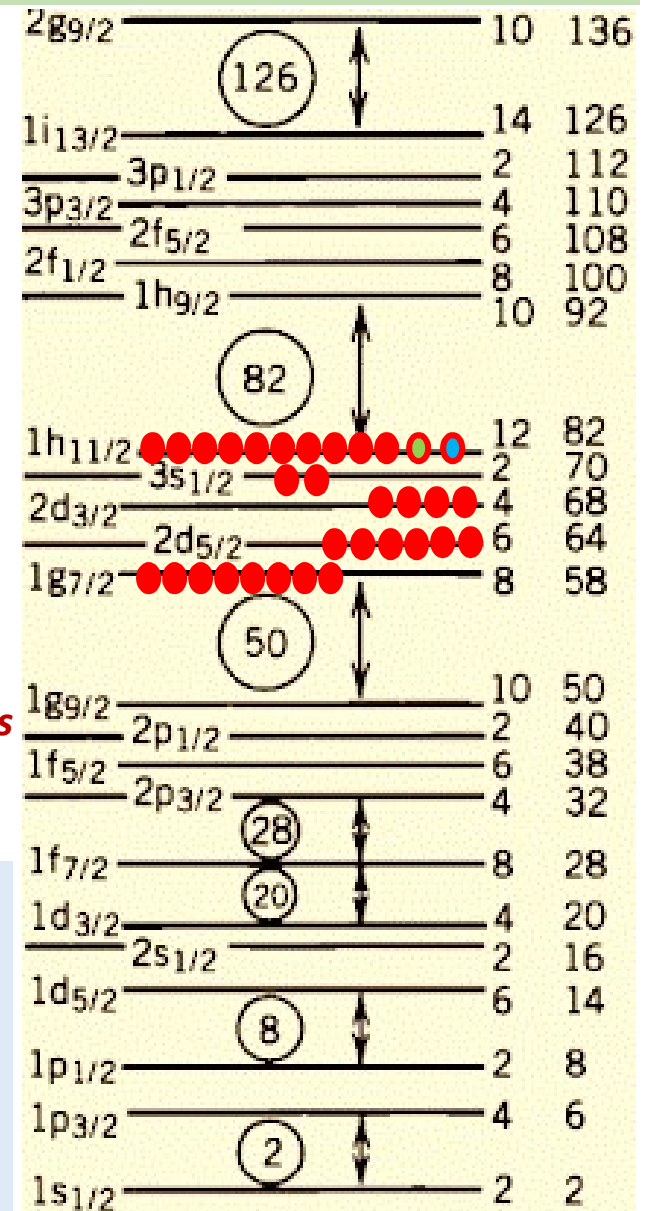
Example

$$j_1 = 2, j_2 = 2$$

$$m_j = m_{j_1} + m_{j_2}$$

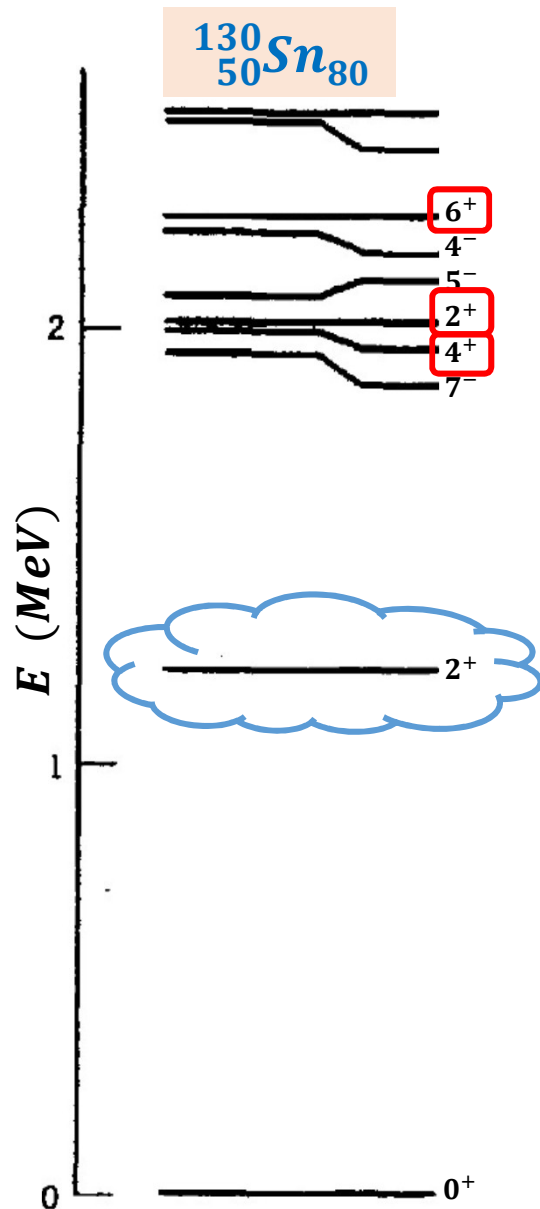
$$j = 0, 2, 4$$

$m_{j_2} \backslash m_{j_1}$	2	1	0	-1	-2
2	4	3	2	1	0
1	3	2	1	0	-1
0	2	1	0	-1	-2
-1	1	0	-1	-2	-3
-2	0	-1	-2	-3	-4



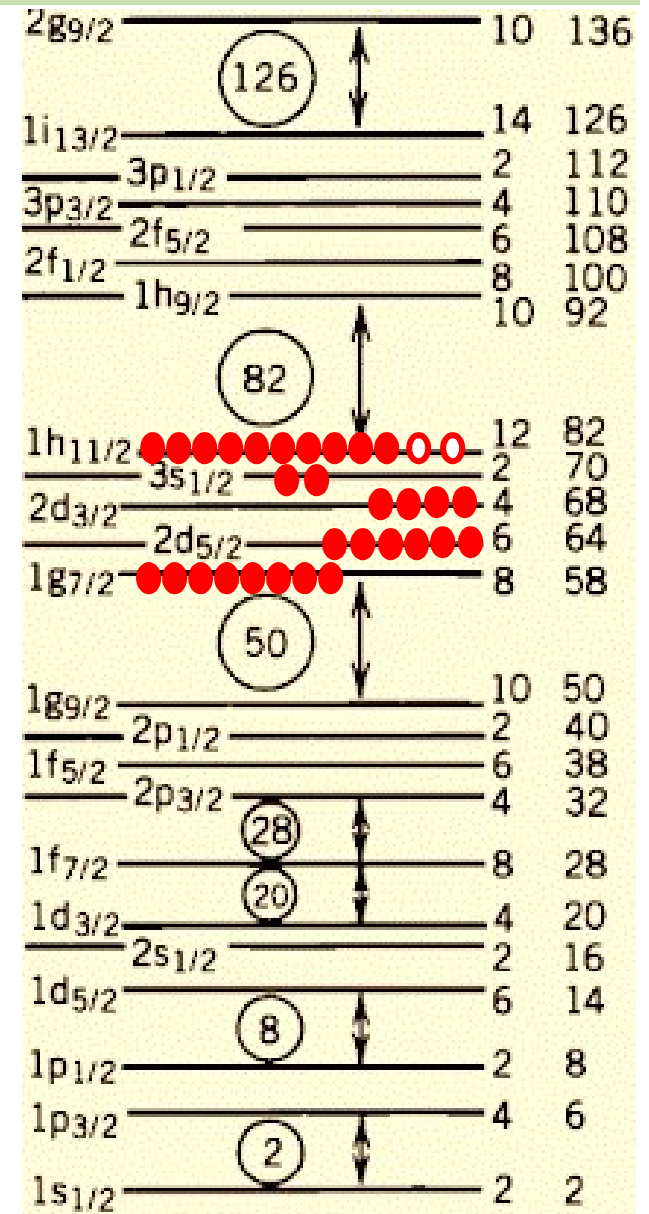
Shell Model → Collective structure

Even-even nuclei



Possibilities

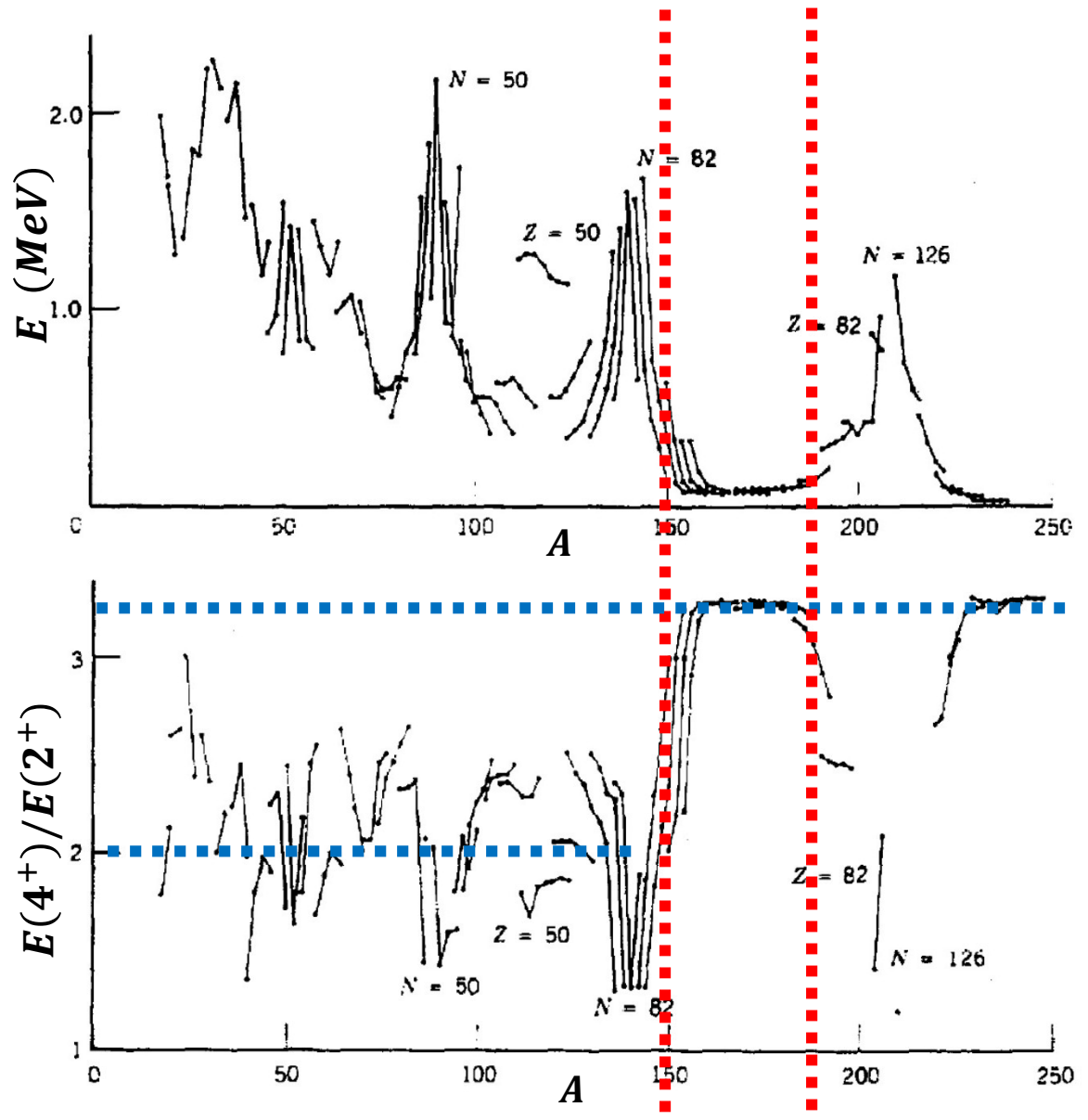
- ☐ No pair breaking and nucleon excitation can give an excited state at this low energy
- ☐ No possible explanation for the first 2^+ state from shell model



Shell Model → Collective structure

Even-even nuclei

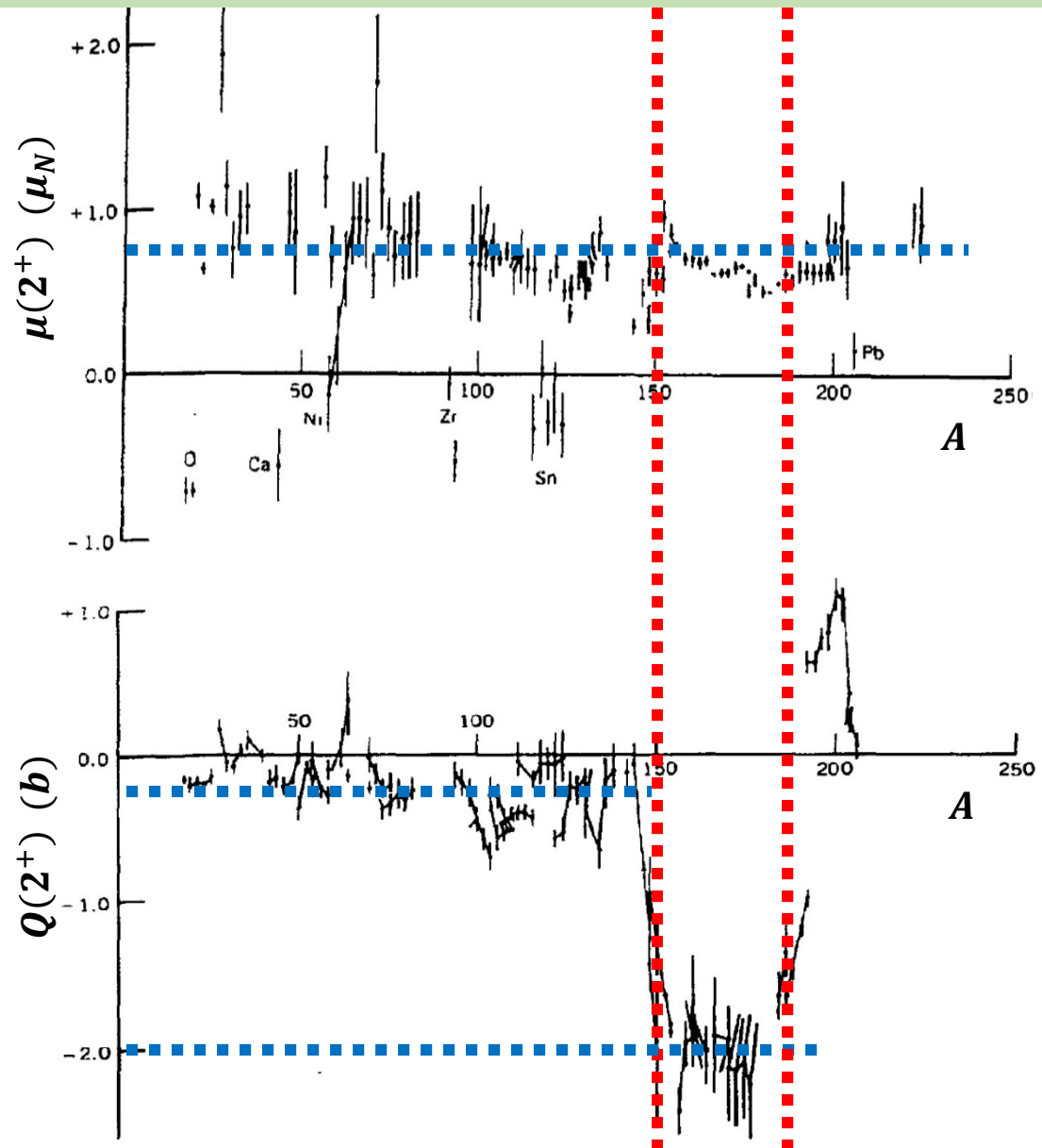
- ❑ *The region $150 < A < 190$ shows values of $E(2^+)$ that are both exceptionally small and remarkably constant*
- ❑ *Except nuclei near closed shells, the ratio $E(4^+)/E(2^+)$ is roughly 2.0 for nuclei $A < 150$ and constant at 3.3 for nuclei $150 < A < 190$ and $A > 230$*



Shell Model → Collective structure

Even-even nuclei

- ❑ *Magnetic moments of the 2^+ states are fairly constant in the range 0.7 – 1.0*
- ❑ *Electric quadrupole moments are small for $A < 150$ and much larger for $A > 150$*
- ❑ *The nuclei with $A < 150$ are generally treated in term of a model based on vibrations about a spherical equilibrium shape*
- ❑ *Nuclei with $150 < A < 190$ show structures most characteristic of rotations of a non-spherical system*
- ❑ *Vibrations & rotations are two major types of collective nuclear motion*



Collective Models

Nuclear Vibrations

For a non-spherical shape,

$$R(t) = R_{av} + \sum_{\lambda=0}^{\infty} \sum_{\mu=-\lambda}^{+\lambda} \alpha_{\lambda\mu}(t) Y_{\lambda}^{\mu}(\theta, \phi)$$

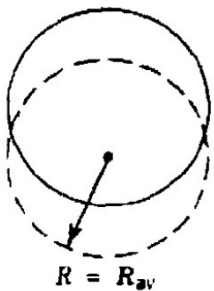
$Y_{\lambda}^{\mu}(\theta, \phi)$: spherical harmonics

Reflection symmetry requires that $\alpha_{\lambda\mu} = \alpha_{\lambda-\mu}$

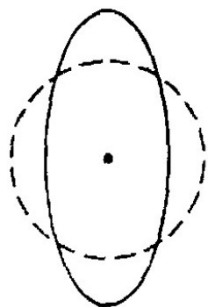
Assuming nuclear fluid is incompressible
further restrictions apply

$$\lambda = 0 \Rightarrow Y_0^0(\theta, \phi) = \sqrt{1/4\pi} = \text{const.}$$

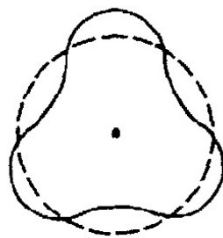
$R(t) = R_{av} + \text{cont.} \Rightarrow \text{no vibrations}$



$\lambda = 1$
(Dipole)



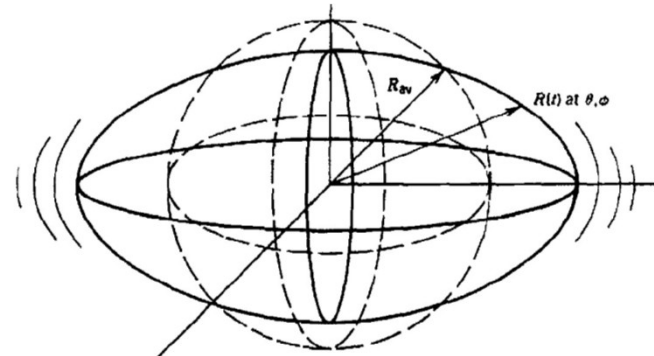
$\lambda = 2$
(Quadrupole)



$\lambda = 3$
(Octupole)

Net displacement of the center of mass $\rightarrow$ Cannot result from internal nuclear forces

Shape vibrations on spherical nucleus



lowest mode of vibration is $\lambda = 2$

- ☐ Vibrations are quantized
- ☐ Energy will be quantized
- ☐ Corresponding to these shape oscillations, we will have
 - $\rightarrow$ energy
 - $\rightarrow$ orbital angular momentum
- ☐ All this will be added to the shell model energy, ang. mom., and parity

Collective models

Nuclear Vibrations

Shape vibrations on spherical nucleus

A quantum of vibration is called phonon

A single unit of $\lambda = 2$ nuclear vibration : quadrupole phonon

(1) Let's consider the effect of adding one unit of vibrational energy (a quadrupole phonon) to the 0^+ ground state of an even-even nucleus

A $\lambda = 2$ phonon carries
✓ 2 units of ang. mom.
✓ Even/positive parity



Adds $Y_{2\mu}$ dependence to
the nuclear wave function
(like Y_{lm} for $l = 2$)

$$\text{parity} = (-1)^l$$

*Adding two units of angular momentum
to a 0^+ state gives only a 2^+ state*



*In exact agreement with observed first
excited state for e-e nuclei*

1 phonon of $\lambda = 2$ type

$$\text{Energy} = \hbar\omega_2$$

Ang. mom. Quantum number, $\lambda = 2$

$$\text{Parity } \pi = +ve$$

$$R(t) = R_{av} + \sum_{\lambda=0}^{\infty} \sum_{\mu=-\lambda}^{+\lambda} \alpha_{\lambda\mu}(t) Y_{\lambda}^{\mu}(\theta, \phi)$$

Collective models

Nuclear Vibrations

Shape vibrations on spherical nucleus

A quantum of vibration is called phonon

A single unit of $\lambda = 2$ nuclear vibration : quadrupole phonon

(2) Let's consider the effect of adding a second quadrupole phonon

2 phonons of $\lambda = 2$ type

$$\text{Energy} = 2\hbar\omega_2$$

$$\text{Ang. mom.}, \lambda = 0, 2, 4$$

$$\text{Parity } \pi = +ve$$

5 possible components μ_1 and μ_2 : 2, 1, 0, -1, -2



25 possible components of $\lambda\mu$
for the two phonons

There is one possible combination with $\mu_{tot} = +4$



Can be associated with a transfer of 4 units of ang. mom.

$$Y_{lm} = Y_{44} \quad (l = 4, m = +4)$$

There are 2 combinations with $\mu_{tot} = +3$

$$\mu_1 = +2 \quad \mu_2 = +1$$

$$\mu_1 = +1 \quad \mu_2 = +2$$

There are 3 combinations with $\mu_{tot} = +2$

$$\mu_1 = +2 \quad \mu_2 = 0 \quad \mu_1 = +1 \quad \mu_2 = +1 \quad \mu_1 = 0 \quad \mu_2 = +2$$

$$\mu_1 = 0 \quad \mu_2 = +1 \quad \mu_1 = -1 \quad \mu_2 = +2$$

$\mu_2 \backslash \mu_1$	2	1	0	-1	-2
2	4	3	2	1	0
1	3	2	1	0	-1
0	2	1	0	-1	-2
-1	1	0	-1	-2	-3
-2	0	-1	-2	-3	-4

Making proper symm. combination of the phonon wavefuncs,
25 combinations reduce to 15 combinations and grouped as

$$\lambda = 4 \quad \mu = +4, +3, +2, +1, 0, -1, -2, -3, -4$$

$$\lambda = 2 \quad \mu = +2, +1, 0, -1, -2$$

$$\lambda = 0 \quad \mu = 0$$

$$R(t) = R_{av} + \sum_{\lambda=0}^{\infty} \sum_{\mu=-\lambda}^{+\lambda} \alpha_{\lambda\mu}(t) Y_{\lambda}^{\mu}(\theta, \phi)$$

Collective models

Nuclear Vibrations

Shape vibrations on spherical nucleus

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$$\text{Ang. mom., } \lambda = 0, 2, 4$$

$$\text{Parity } \pi = +ve$$

5 possible components μ_1 and μ_2 : 2, 1, 0, -1, -2



25 possible components of $\lambda\mu$
for the two phonons

$\mu_2 \backslash \mu_1$		2	1	0	-1	-2
$\mu_{tot} = \mu_1 + \mu_2$	2	4	3	2	1	0
	1	3	2	1	0	-1
	0	2	1	0	-1	-2
	-1	1	0	-1	-2	-3
	-2	0	-1	-2	-3	-4

There is one possible combination with $\mu_{tot} = +4$



Can be associated with a transfer of 4 units of ang. mom.

$$Y_{lm} = Y_{44} \quad (l = 4, m = +4)$$

There are 2 combinations with $\mu_{tot} = +3$

$$\mu_1 = +2 \quad \mu_2 = +1$$

$$\mu_1 = +1 \quad \mu_2 = +2$$

There are 3 combinations with $\mu_{tot} = +2$

$$\mu_1 = +2 \quad \mu_2 = 0 \quad \mu_1 = +1 \quad \mu_2 = +1 \quad \mu_1 = 0 \quad \mu_2 = +2$$

$$\mu_1 = 0 \quad \mu_2 = +1 \quad \mu_1 = +1 \quad \mu_2 = 0 \quad \mu_1 = +2 \quad \mu_2 = -1$$

Making proper symm. combination of the phonon wavefuncs,
25 combinations reduce to 15 combinations and grouped as

$$\lambda = 4 \quad \mu = +4, +3, +2, +1, 0, -1, -2, -3, -4$$

$$\lambda = 2 \quad \mu = +2, +1, 0, -1, -2$$

$$\lambda = 0 \quad \mu = 0$$

We expect a triplet of states with spins
 $0^+, 2^+, 4^+$ at twice the energy of the
first 2^+ state (2 phonons)

Collective models

Nuclear Vibrations

Shape vibrations on spherical nucleus

A quantum of vibration is called phonon

A single unit of $\lambda = 2$ nuclear vibration : quadrupole phonon

1 phonon of $\lambda = 2$ type

$$\text{Energy} = \hbar\omega_2$$

Ang. mom. Quantum number, $\lambda = 2$

$$\text{Parity } \pi = +ve$$

2 phonons of $\lambda = 2$ type

$$\text{Energy} = 2\hbar\omega_2$$

Ang. mom., $\lambda = 0, 2, 4$

$$\text{Parity } \pi = +ve$$

3 phonon of $\lambda = 2$ type

$$\text{Energy} = 3\hbar\omega_2$$

Ang. mom. $\lambda = 0, 2, 3, 4, 6$

$$\text{Parity } \pi = +ve$$

1 phonon of $\lambda = 3$ type

$$\text{Energy} = \hbar\omega_3$$

Ang. mom. Quantum number, $\lambda = 3$

$$\text{Parity } \pi = -ve$$

Energy of 1 phonon state of $\lambda = 3$ type $\sim$ Energy of 2 phonon states of $\lambda = 2$ type

The 1st 2^+ state in e-e nuclei can come from 1 phonon state of $\lambda = 2$ type

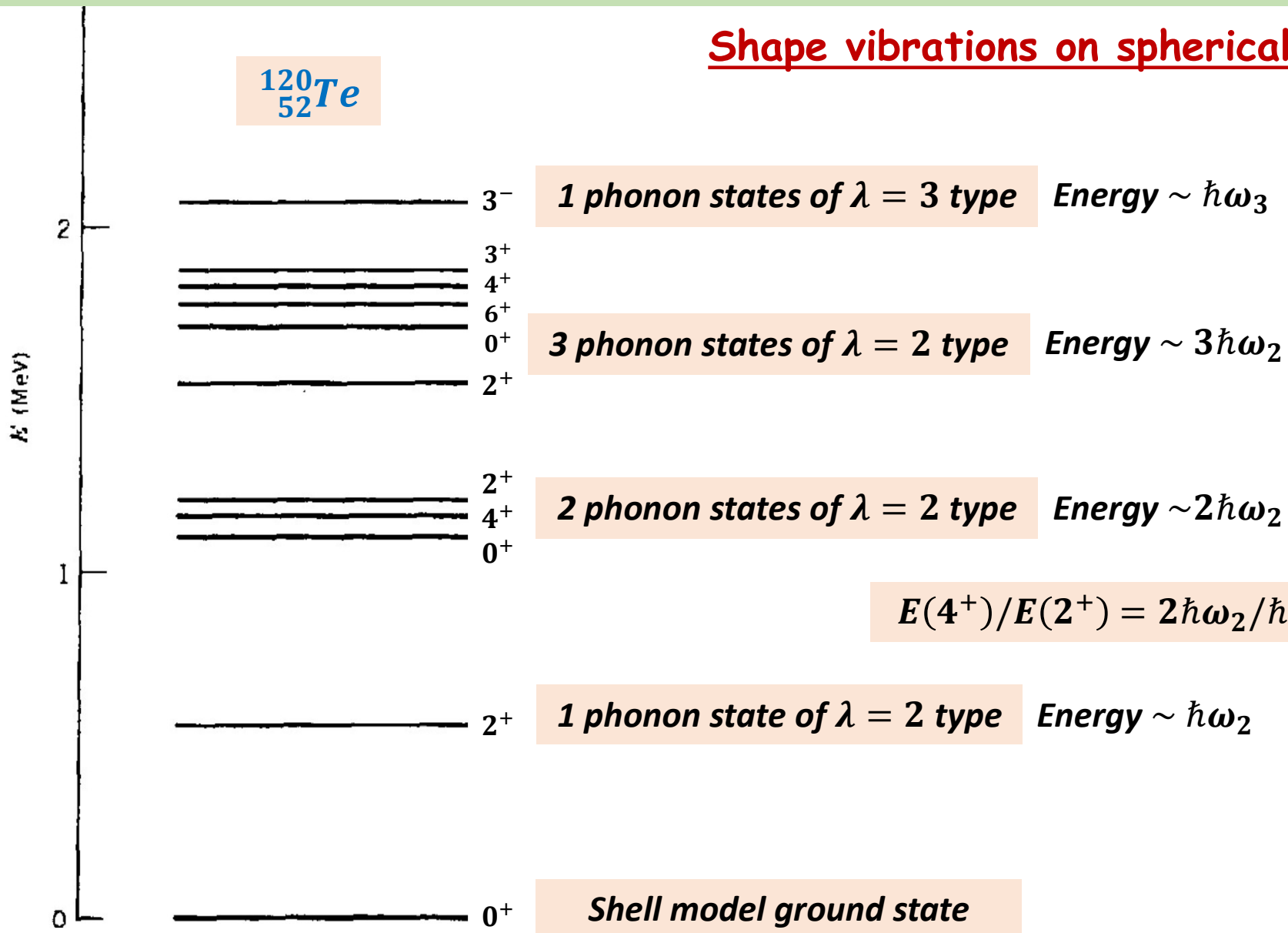
The 1st 4^+ state in e-e nuclei can come from 2 phonon state of $\lambda = 2$ type

Collective models

Nuclear Vibrations

Shape vibrations on spherical nucleus

$^{120}_{52}\text{Te}$

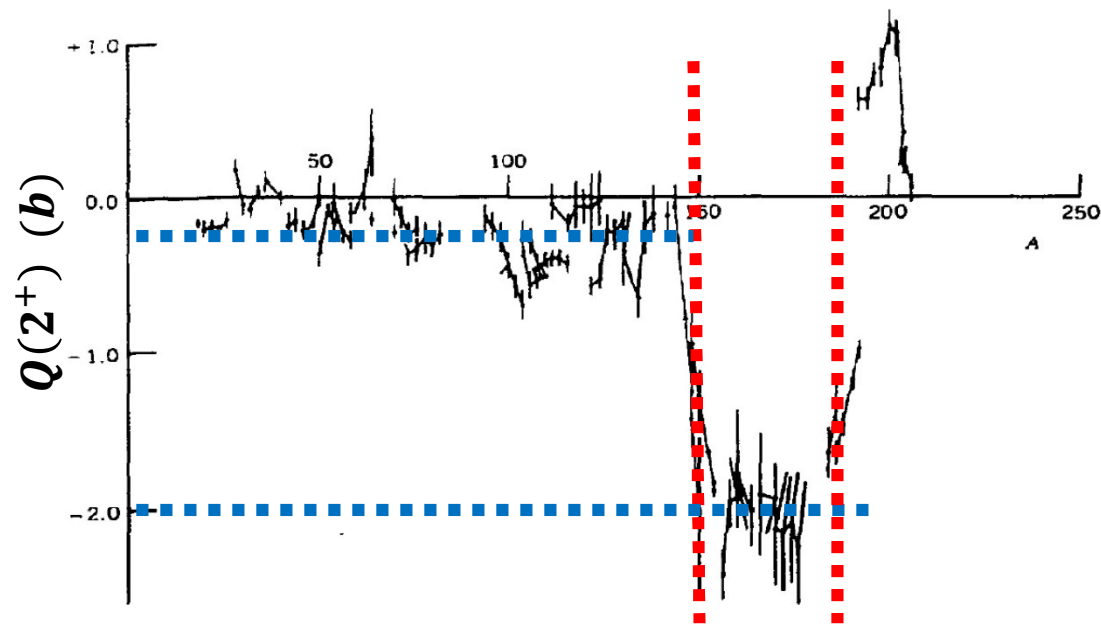


Collective models

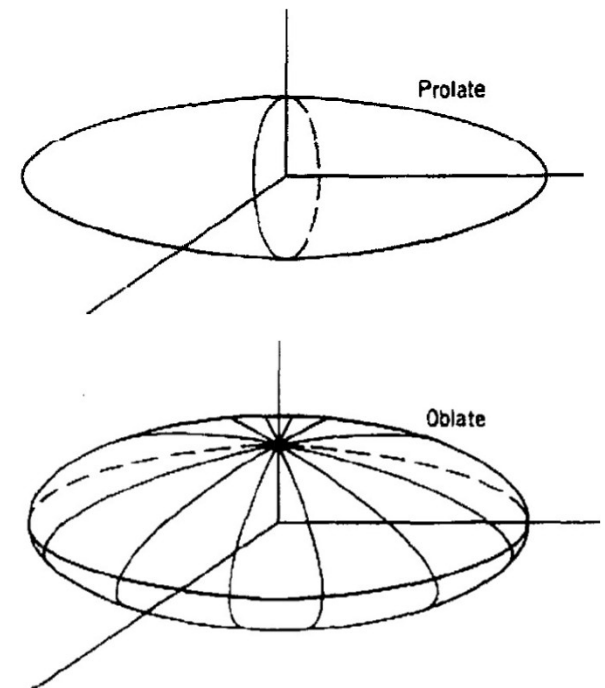
Nuclear Rotation

Deformed Nuclei

Rotational motion can be observed only in nuclei with *non-spherical equilibrium shapes*



Large quadrupole moment



$$R(\theta, \phi) = R_{av} [1 + \beta Y_2^0(\theta, \phi)]$$

$$\beta = \frac{4}{3} \sqrt{\frac{\pi}{5}} \frac{\Delta R}{R_{av}}$$

$$Q_0 = \frac{3}{\sqrt{5\pi}} R_{av}^2 Z \beta (1 + 0.16 \beta)$$

$$R_{av} = R_0 A^{1/3}$$

β : deformation parameter

ΔR : diff. betⁿ semi-major and semi-minor axis of the ellipse

Rotation about an axis perpendicular to the symmetry axis $\Rightarrow$ rotational band

Collective models

Nuclear Rotation

$$E = \frac{1}{2} \mathcal{I} \omega^2 \quad L = \mathcal{I} \omega \quad L^2 = l(l+1) \hbar^2$$

$$= \frac{1}{2} \mathcal{I} \frac{L^2}{\mathcal{I}^2} = \frac{L^2}{2\mathcal{I}} = \frac{\hbar^2}{2\mathcal{I}} l(l+1)$$

$$L \rightarrow I$$

$$E = \frac{\hbar^2}{2\mathcal{I}} I(I+1)$$

$$E(0^+) = 0$$

$$E(2^+) = \frac{\hbar^2}{2\mathcal{I}} 6$$

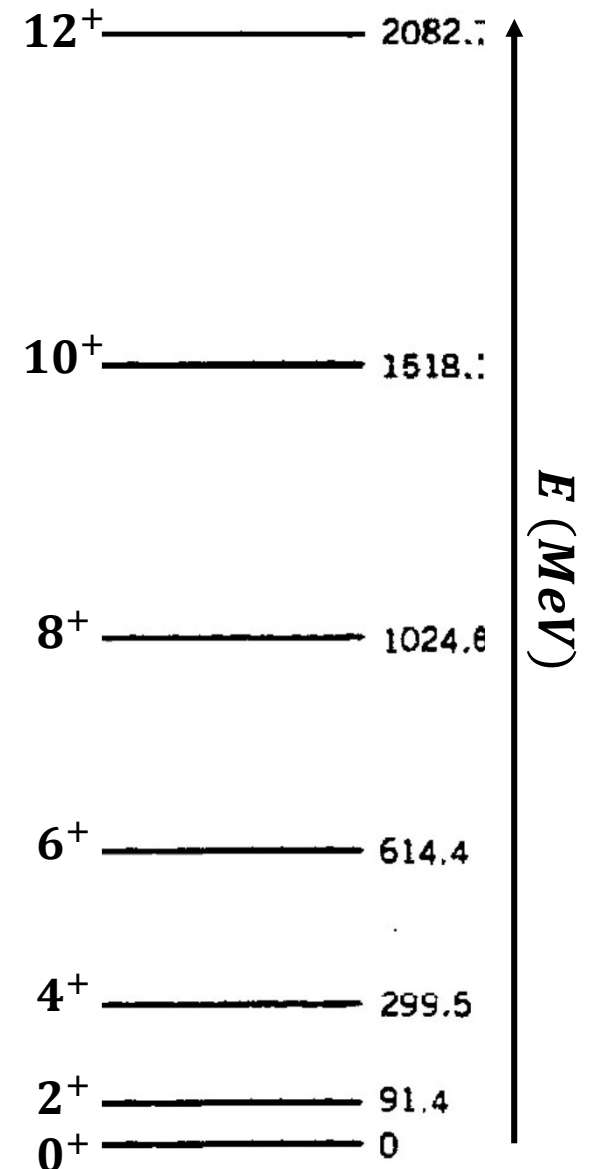
$$E(4^+) = \frac{\hbar^2}{2\mathcal{I}} 20$$

$$E(6^+) = \frac{\hbar^2}{2\mathcal{I}} 42$$

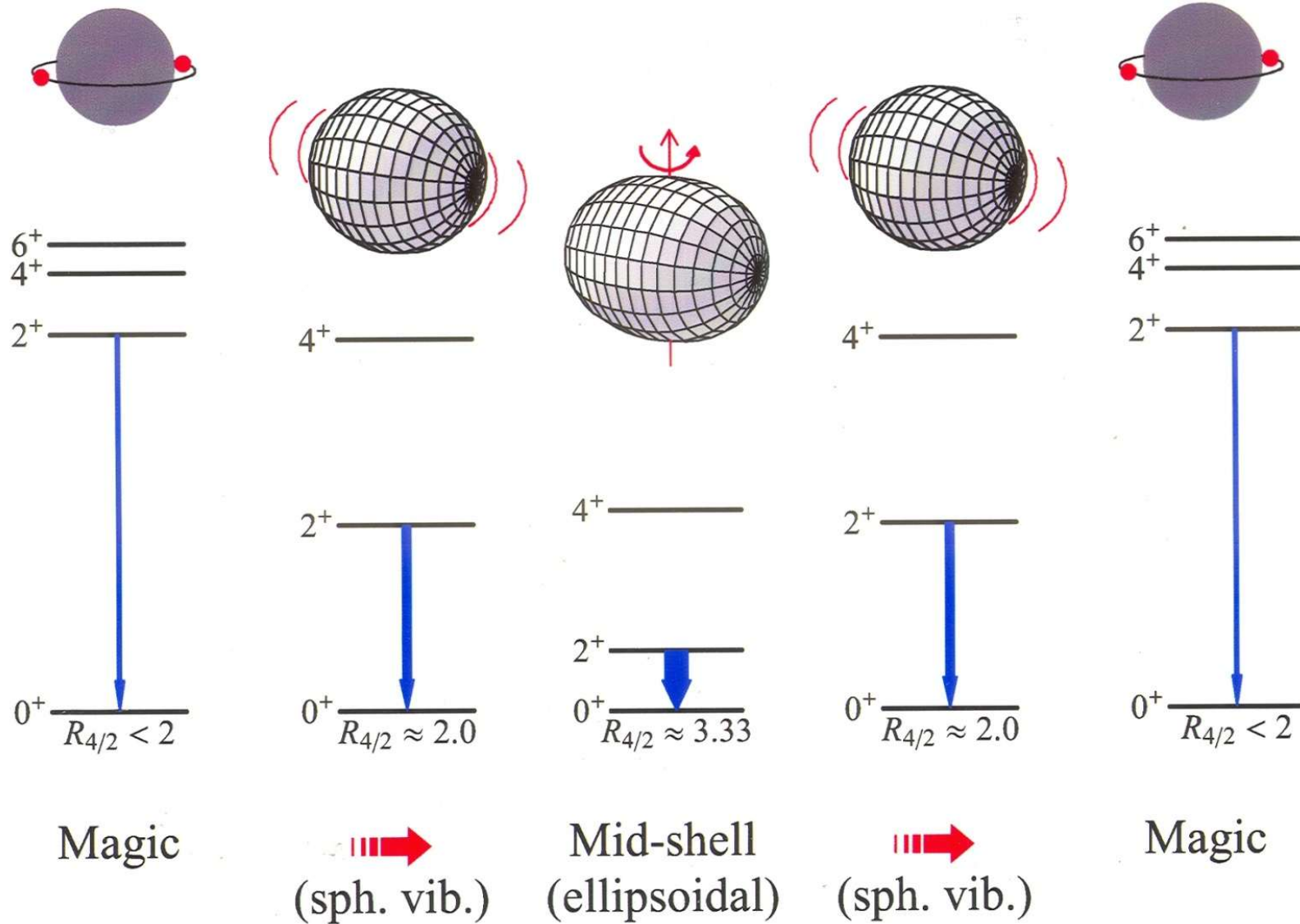
$$E(8^+) = \frac{\hbar^2}{2\mathcal{I}} 72$$

**Rotational
band
appears**

Mirror symmetry $\Rightarrow I = \text{even}$



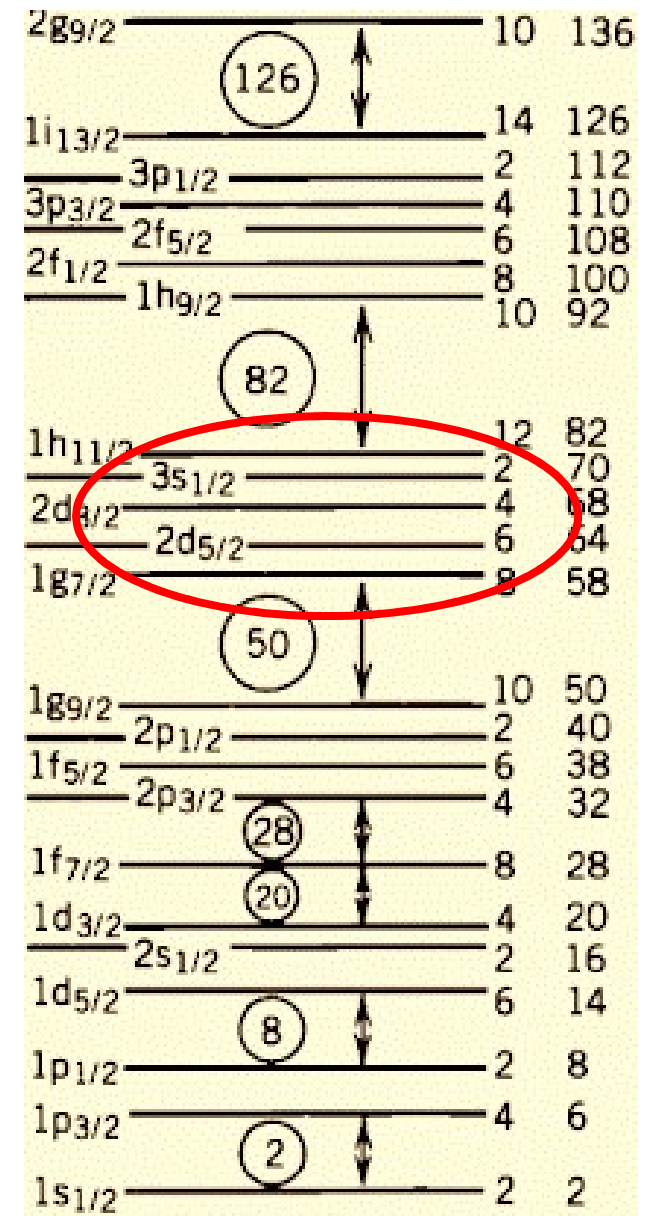
Evolution of nuclear structure as a func. of nucleon no.



More realistic nuclear Models

□ Many-body shell model

□ Configuration mixing



More realistic nuclear Models

□ Single particle states in deformed nuclei

Nilsson model

